MEDICAL KNOWLEDGE BASE

Comprehensive Symptom & Disease Reference Guide

Optimized for RAG-Based Medical Chatbot Applications

• 16 Medical Domains • 60+ Diseases • Symptoms, Causes, Diagnosis, Treatment

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16	60+	500+	100%
Medical Domains	Disease Profiles	Symptoms Mapped	PyPDF Compatible

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Introduction & How to Use This Document

This Medical Knowledge Base is a comprehensive, structured reference document designed specifically to serve as a high-quality corpus for Retrieval-Augmented Generation (RAG) systems powering medical chatbots and clinical decision support tools.

The document covers 16 major medical domains and over 60 disease profiles, each structured with consistent sections to maximize embedding quality and retrieval accuracy. Every disease entry follows the same format: Overview, Symptoms, Causes, Risk Factors, Diagnosis, Treatment, Prevention, and When to See a Doctor.

Each section has been written to be semantically rich and self-contained, ensuring that vector embeddings generated from individual sections carry sufficient contextual information for accurate retrieval. The consistent structure also facilitates metadata-based filtering in hybrid retrieval systems.

IMPORTANT DISCLAIMER: This document is intended solely for educational, research, and AI development purposes. It does not constitute medical advice. All medical decisions must be made in consultation with qualified healthcare professionals. The information herein, while carefully compiled, may not reflect the most current clinical guidelines.

Section	Content	RAG Use Case
Overview	Disease definition, epidemiology, global burden	General disease queries
Symptoms	Primary and secondary clinical features	Symptom-to-disease matching
Causes	Etiology and pathophysiology	Mechanism queries
Risk Factors	Predisposing conditions and demographics	Risk assessment queries
Diagnosis	Diagnostic criteria, investigations, tests	Diagnostic decision support
Treatment	Pharmacological and non-pharmacological management	Treatment recommendation
Prevention	Preventive strategies and public health measures	Prevention advice queries
When to See a Doctor	Red flags and emergency indicators	Urgency triage queries

General Medicine

Influenza (Flu) J11

Overview

Influenza is a contagious respiratory illness caused by influenza viruses. It can cause mild to severe illness and can lead to hospitalization or death, particularly in high-risk groups.

Symptoms & Clinical Features

• High fever (38–40 C)	• Sore throat
• Severe headache	• Runny or stuffy nose
• Muscle and body aches	• Chills
• Fatigue and weakness	• Loss of appetite
• Dry cough	• Occasional vomiting/diarrhea (more common in children)

Causes & Pathophysiology

Influenza A and B viruses transmitted via respiratory droplets from coughing, sneezing, or talking. Also spreads via surfaces.

Risk Factors

- Age > 65 or < 5 years
- Pregnancy
- Chronic illnesses (asthma, diabetes, heart disease)
- Immunosuppression
- Obesity (BMI > 40)

Diagnosis

Clinical presentation; confirmed by rapid influenza diagnostic tests (RIDTs), RT-PCR, or viral culture.

Treatment & Management

- Antiviral medications: Oseltamivir (Tamiflu), Zanamivir, Baloxavir within 48 hours of symptom onset
- Rest and adequate hydration
- Antipyretics: Paracetamol or Ibuprofen for fever
- Avoid aspirin in children (risk of Reye syndrome)

Prevention

- Annual influenza vaccination
- Frequent handwashing
- Avoid close contact with sick individuals
- Cover mouth and nose when coughing/sneezing

When to See a Doctor

■ **Seek immediate care for difficulty breathing, persistent chest pain, confusion, severe vomiting, or bluish lips.**

Type 2 Diabetes Mellitus E11

Overview

A chronic metabolic disorder characterized by high blood glucose levels resulting from insulin resistance and relative insulin deficiency. It is the most prevalent form of diabetes, accounting for 90-95% of all cases.

Symptoms & Clinical Features

- Increased thirst (polydipsia)
- Frequent urination (polyuria)
- Unexplained weight loss
- Fatigue
- Blurred vision
- Slow-healing sores or frequent infections
- Tingling or numbness in hands/feet
- Darkened skin in body creases (acanthosis nigricans)

Causes & Pathophysiology

Insulin resistance in muscle, fat, and liver cells, combined with progressive beta-cell failure. Strongly associated with obesity and sedentary lifestyle.

Risk Factors

- Overweight or obesity
- Physical inactivity
- Family history
- Prediabetes
- Gestational diabetes history
- Polycystic ovarian syndrome
- Age > 45

Diagnosis

Fasting plasma glucose >= 126 mg/dL, HbA1c >= 6.5%, 2-hour plasma glucose >= 200 mg/dL during OGTT, or random glucose >= 200 mg/dL with symptoms.

Treatment & Management

- Lifestyle modification: diet and exercise
- Metformin (first-line oral agent)
- SGLT-2 inhibitors, GLP-1 agonists, DPP-4 inhibitors, sulfonylureas
- Insulin therapy when oral agents are insufficient
- Blood pressure and lipid management

Prevention

- Maintain healthy weight
- Regular physical activity (150 min/week)
- Healthy diet rich in fiber
- Regular blood glucose monitoring in high-risk individuals

When to See a Doctor

■ Immediately if experiencing symptoms of hyperglycemia (extreme thirst, confusion) or hypoglycemia (shakiness, sweating, loss of consciousness).

Hypertension (High Blood Pressure) I10

Overview

A chronic condition in which the blood pressure in the arteries is persistently elevated (>=140/90 mmHg). Often called the 'silent killer' because it typically has no symptoms until serious complications occur.

Symptoms & Clinical Features

- Usually asymptomatic
- Headache (especially in morning)
- Dizziness
- Blurred vision
- Shortness of breath
- Nosebleeds (in severe cases)
- Chest pain (hypertensive crisis)

Causes & Pathophysiology

Primary (essential) hypertension has no single cause. Secondary hypertension is caused by kidney disease, adrenal disorders, thyroid problems, or certain medications.

Risk Factors

- Age (risk increases with age)
- Family history
- Obesity
- Sedentary lifestyle
- High-sodium diet
- Smoking
- Excessive alcohol
- Stress
- Diabetes
- Chronic kidney disease

Diagnosis

Blood pressure measurement >= 140/90 mmHg on two or more separate occasions. Includes ambulatory blood pressure monitoring and home measurements.

Treatment & Management

- ACE inhibitors, ARBs, calcium channel blockers, diuretics, beta-blockers
- Lifestyle modifications: DASH diet, weight loss, exercise
- Sodium restriction < 2g/day
- Limit alcohol intake
- Smoking cessation

Prevention

- Regular blood pressure checks
- Healthy diet low in sodium
- Regular exercise
- Maintain healthy weight
- Limit alcohol and caffeine
- Stress management

When to See a Doctor

■ **Seek emergency care if BP > 180/120 mmHg with symptoms of organ damage (chest pain, vision changes, numbness, difficulty speaking).**

Common Cold (Upper Respiratory Tract Infection) J00

Overview

A viral infectious disease of the upper respiratory tract affecting primarily the nose. It is the most common infectious disease in humans, with adults experiencing 2-3 colds per year on average.

Symptoms & Clinical Features

• Runny or stuffy nose	• Mild headache
• Sore throat	• Body aches
• Cough	• Watery eyes
• Sneezing	• Mild fatigue
• Low-grade fever	• Loss of smell/taste

Causes & Pathophysiology

Over 200 virus strains, most commonly rhinoviruses (50%). Also caused by coronaviruses, adenoviruses, and respiratory syncytial virus (RSV).

Risk Factors

- Age (children more susceptible)
- Weakened immune system
- Exposure to cold weather
- Crowded environments
- Smoking

Diagnosis

Clinical diagnosis based on symptoms. No specific tests required unless ruling out influenza or COVID-19.

Treatment & Management

- Symptomatic: rest, fluids, and over-the-counter decongestants
- Saline nasal rinse
- Throat lozenges
- Paracetamol for pain and fever
- No antibiotics (viral etiology)
- Honey for cough in adults and children > 1 year

Prevention

- Frequent handwashing
- Avoid touching face
- Avoid close contact with infected individuals
- Disinfect surfaces

When to See a Doctor

■ **If symptoms worsen after 10 days, fever > 39.4 C, severe headache, or difficulty breathing.**

Respiratory Medicine

Asthma J45

Overview

A chronic inflammatory disease of the airways characterized by variable and recurring symptoms, reversible airflow obstruction, and bronchospasm. Affects approximately 300 million people worldwide.

Symptoms & Clinical Features

- Wheezing
- Shortness of breath
- Chest tightness
- Cough (worse at night or early morning)
- Difficulty sleeping due to breathing problems
- Rapid breathing
- Fatigue during exercise

Causes & Pathophysiology

Combination of genetic predisposition and environmental factors causing airway inflammation. Triggers include allergens, cold air, exercise, respiratory infections, smoke, and stress.

Risk Factors

- Family history of asthma or allergies
- Atopic dermatitis or allergic rhinitis
- Exposure to tobacco smoke
- Air pollution
- Occupational exposures
- Obesity

Diagnosis

Spirometry showing FEV1/FVC ratio < 0.70 with bronchodilator reversibility (> 12% and 200mL improvement in FEV1). Peak flow monitoring, methacholine challenge test.

Treatment & Management

- Short-acting beta-agonists (SABA): Salbutamol for acute relief
- Inhaled corticosteroids (ICS): Budesonide, Beclomethasone for long-term control
- Long-acting beta-agonists (LABA): Formoterol, Salmeterol
- Leukotriene receptor antagonists: Montelukast
- Biologics for severe asthma: Omalizumab, Mepolizumab

Prevention

- Identify and avoid triggers
- Use peak flow meter to monitor lung function
- Take controller medications as prescribed
- Get annual flu vaccine
- Maintain healthy weight

When to See a Doctor

■ **Emergency care if severe breathlessness, inability to speak, bluish lips, no improvement with rescue inhaler.**

Chronic Obstructive Pulmonary Disease (COPD) J44

Overview

A progressive inflammatory lung disease causing obstructed airflow. Encompasses chronic bronchitis and emphysema. Third leading cause of death worldwide, almost exclusively caused by tobacco smoking.

Symptoms & Clinical Features

- Chronic productive cough
- Dyspnea (shortness of breath)
- Wheezing
- Chest tightness
- Frequent respiratory infections
- Cyanosis (bluish discoloration of lips)
- Barrel chest
- Weight loss in advanced disease

Causes & Pathophysiology

Long-term exposure to irritating gases or particulate matter, most often cigarette smoke. Also caused by air pollution, occupational dust and chemicals.

Risk Factors

- Cigarette smoking (primary cause in 85-90%)
- Long-term occupational exposure to dust/chemicals
- Indoor air pollution (biomass fuels)
- Alpha-1 antitrypsin deficiency (genetic)
- Age > 40

Diagnosis

Post-bronchodilator spirometry: FEV1/FVC < 0.70. GOLD staging based on FEV1% predicted. Chest X-ray, CT scan, arterial blood gases.

Treatment & Management

- Smoking cessation (most important intervention)
- Short-acting bronchodilators: SABA, SAMA (Ipratropium)
- Long-acting bronchodilators: LABA, LAMA (Tiotropium)
- Inhaled corticosteroids for frequent exacerbations
- Pulmonary rehabilitation
- Oxygen therapy for hypoxemia
- Lung volume reduction surgery or transplantation in selected cases

Prevention

- Never start smoking; quit if smoker
- Avoid occupational and environmental exposures
- Annual influenza and pneumococcal vaccination

When to See a Doctor

■ **Emergency care for sudden severe breathlessness, confusion, or cyanosis.**

Pneumonia J18

Overview

An infection that inflames the air sacs in one or both lungs. The air sacs may fill with fluid or pus, causing cough with phlegm or pus, fever, chills, and difficulty breathing. Leading infectious cause of death worldwide.

Symptoms & Clinical Features

• High fever with chills	• Rapid heart rate
• Productive cough with rust-colored or greenish sputum	• Fatigue
• Pleuritic chest pain	• Nausea and vomiting
• Shortness of breath	• Confusion (especially elderly)
• Rapid breathing (tachypnea)	• Low oxygen saturation

Causes & Pathophysiology

Bacterial (Streptococcus pneumoniae most common), viral (influenza, COVID-19, RSV), fungal (Pneumocystis jirovecii in immunocompromised). Aspiration pneumonia from inhaling food/liquid.

Risk Factors

- Age < 2 or > 65
- Smoking
- Chronic lung disease
- Weakened immune system
- Hospitalization or mechanical ventilation
- Swallowing difficulties

Diagnosis

Chest X-ray or CT showing consolidation, infiltrates. Sputum culture and sensitivity. Blood cultures. Complete blood count (elevated WBC). Urinary antigen tests for Legionella and Streptococcus.

Treatment & Management

- Antibiotics based on pathogen: Amoxicillin, Azithromycin, Levofloxacin
- Antivirals for viral pneumonia
- Supplemental oxygen
- IV fluids
- Antipyretics
- Hospitalization for severe cases (CURB-65 score)

Prevention

- Pneumococcal and influenza vaccination
- Good hygiene
- Smoking cessation
- Adequate nutrition

When to See a Doctor

■ **Immediately for high fever, rapid breathing, chest pain, confusion, or oxygen saturation < 94%.**

Pulmonary Tuberculosis A15

Overview

A potentially serious infectious disease mainly affecting the lungs caused by Mycobacterium tuberculosis. One of the top 10 causes of death worldwide and the leading cause from a single infectious agent.

Symptoms & Clinical Features

- Persistent cough lasting more than 3 weeks
- Coughing up blood (hemoptysis)
- Chest pain
- Unintentional weight loss
- Night sweats
- Fever
- Fatigue
- Loss of appetite
- Lymph node enlargement

Causes & Pathophysiology

Mycobacterium tuberculosis spread through airborne droplets from person with active TB. Not all infected develop disease (latent vs. active TB).

Risk Factors

- HIV infection
- Malnutrition
- Diabetes
- Close contact with active TB case
- Living in high-prevalence areas
- Immunosuppressive medications

Diagnosis

Sputum smear microscopy, culture (gold standard), GeneXpert MTB/RIF (rapid molecular test), tuberculin skin test (TST), IGRA blood test, chest X-ray.

Treatment & Management

- Standard DOTS regimen: Isoniazid + Rifampicin + Pyrazinamide + Ethambutol for 2 months, followed by Isoniazid + Rifampicin for 4 months
- Drug-resistant TB requires individualized longer regimens
- Directly Observed Therapy (DOT) to ensure adherence

Prevention

- BCG vaccination in infants
- Early diagnosis and treatment
- Infection control measures
- Treatment of latent TB in high-risk individuals

When to See a Doctor

■ **Seek care if cough persists > 3 weeks, especially with blood, night sweats, or unexplained weight loss.**

Cardiovascular Medicine

Acute Myocardial Infarction (Heart Attack) I21

Overview

Occurs when blood flow to part of the heart muscle is blocked, causing irreversible ischemic damage. A leading cause of death globally, requiring immediate medical attention.

Symptoms & Clinical Features

- Crushing chest pain or pressure radiating to arm, jaw, back
- Shortness of breath
- Cold sweats
- Nausea and vomiting
- Lightheadedness or dizziness
- Fatigue
- Palpitations
- Silent MI may be asymptomatic (especially in diabetics and women)

Causes & Pathophysiology

Rupture of atherosclerotic plaque with subsequent thrombus formation occluding a coronary artery. Less commonly: coronary artery spasm, embolism, or dissection.

Risk Factors

- Hypertension
- Hyperlipidemia
- Diabetes
- Smoking
- Obesity
- Sedentary lifestyle
- Family history of CAD
- Age (men > 45, women > 55)
- Stress

Diagnosis

12-lead ECG (ST elevation or depression, T-wave changes), cardiac biomarkers (Troponin I and T, CKMB), echocardiography, coronary angiography.

Treatment & Management

- Immediate aspirin 300mg + P2Y12 inhibitor (Clopidogrel/Ticagrelor)
- Primary PCI (percutaneous coronary intervention) within 90 minutes for STEMI
- Fibrinolysis if PCI not available within 120 min
- Anticoagulation: Heparin, Enoxaparin
- Beta-blockers, ACE inhibitors, statins for long-term
- Cardiac rehabilitation

Prevention

- Control of risk factors: BP, cholesterol, diabetes
- Smoking cessation
- Regular exercise
- Healthy diet
- Low-dose aspirin in selected high-risk patients

When to See a Doctor

■ EMERGENCY — Call ambulance immediately. Time is muscle: every minute of delay causes irreversible heart damage.

Heart Failure I50

Overview

A clinical syndrome where the heart cannot pump enough blood to meet the body's demands. Affects over 64 million people worldwide. Can be systolic (reduced ejection fraction) or diastolic (preserved ejection fraction).

Symptoms & Clinical Features

- Dyspnea on exertion
- Orthopnea (breathlessness when lying flat)
- Paroxysmal nocturnal dyspnea
- Bilateral ankle and leg edema
- Fatigue and weakness
- Rapid or irregular heartbeat
- Persistent cough or wheezing
- Abdominal bloating and nausea
- Rapid weight gain from fluid retention

Causes & Pathophysiology

Coronary artery disease, hypertension, cardiomyopathy, valvular disease, arrhythmias, congenital heart defects, myocarditis.

Risk Factors

- Prior myocardial infarction
- Hypertension
- Diabetes
- Obesity
- Kidney disease
- Alcohol abuse
- Sleep apnea
- Age > 65

Diagnosis

Clinical assessment, BNP/NT-proBNP blood test, echocardiography (EF < 40% for HFrEF), chest X-ray (cardiomegaly, pulmonary congestion), ECG.

Treatment & Management

- ACE inhibitors/ARBs and beta-blockers (cornerstone therapy)
- Aldosterone antagonists: Spironolactone
- SGLT-2 inhibitors: Dapagliflozin, Empagliflozin
- Loop diuretics for fluid management: Furosemide
- ICD/CRT devices for eligible patients

- Heart transplantation in end-stage disease

Prevention

- Aggressive management of hypertension and coronary artery disease
- Limit alcohol intake
- Regular exercise
- Sodium and fluid restriction

When to See a Doctor

■ Seek emergency care for sudden severe breathlessness, chest pain, syncope, or rapid weight gain (> 2kg in 2 days).

Atrial Fibrillation I48

Overview

The most common sustained cardiac arrhythmia, characterized by disorganized atrial electrical activity causing an irregular and often rapid heart rate. Significantly increases risk of stroke and heart failure.

Symptoms & Clinical Features

- Palpitations (rapid, irregular heartbeat)
- Chest pain or discomfort
- Shortness of breath
- Reduced exercise tolerance
- Fatigue
- May be completely asymptomatic
- Dizziness or lightheadedness

Causes & Pathophysiology

Structural heart disease, hypertension, heart failure, valvular disease, thyroid dysfunction, excessive alcohol (holiday heart syndrome), sleep apnea, pulmonary disease.

Risk Factors

- Age > 60
- Hypertension
- Heart disease
- Obesity
- Hyperthyroidism
- Heavy alcohol use
- Obstructive sleep apnea
- European ancestry

Diagnosis

12-lead ECG showing absent P waves with irregularly irregular rhythm. 24-hour Holter monitor for paroxysmal AF. Echocardiography, thyroid function tests.

Treatment & Management

- Rate control: Beta-blockers, calcium channel blockers, Digoxin
- Rhythm control: Cardioversion, antiarrhythmic drugs (Amiodarone, Flecainide)
- Anticoagulation to prevent stroke: Warfarin or DOACs (Apixaban, Rivaroxaban)
- Catheter ablation for recurrent symptomatic AF

- CHA2DS2-VASc score to determine stroke risk

Prevention

- Control of risk factors (hypertension, obesity)
- Limit alcohol intake
- Treat sleep apnea
- Regular physical activity

When to See a Doctor

■ Immediately for new-onset palpitations, chest pain, breathlessness, or signs of stroke (FAST: Face, Arms, Speech, Time).

Neurology

Stroke (Cerebrovascular Accident) I63/I64

Overview

A medical emergency occurring when blood supply to part of the brain is interrupted (ischemic stroke, 87%) or a blood vessel in the brain ruptures (hemorrhagic stroke). Leading cause of disability worldwide.

Symptoms & Clinical Features

- FAST: Face drooping, Arm weakness, Speech difficulty, Time to call emergency
- Sudden numbness or weakness of face/arm/leg (usually one side)
- Sudden severe headache 'thunderclap'
- Sudden confusion or trouble understanding
- Sudden vision changes in one or both eyes
- Sudden dizziness, loss of balance or coordination

Causes & Pathophysiology

Ischemic: thrombosis (clot in cerebral artery) or embolism (clot traveling from heart/carotid). Hemorrhagic: hypertension-induced rupture, aneurysm rupture, AVM.

Risk Factors

- Hypertension (most important modifiable risk factor)
- Atrial fibrillation
- Diabetes
- High cholesterol
- Smoking
- Obesity
- Physical inactivity
- Prior TIA or stroke
- Age > 55
- Male sex

Diagnosis

Urgent CT scan (rules out hemorrhage), MRI-DWI (detects ischemia within minutes), CT/MR angiography, carotid Doppler, ECG, echocardiogram.

Treatment & Management

- Ischemic: IV alteplase (tPA) within 4.5 hours, mechanical thrombectomy within 24 hours
- Hemorrhagic: blood pressure control, reversal of anticoagulation, surgical evacuation if indicated
- Aspirin + clopidogrel for minor stroke/TIA
- Anticoagulation for cardioembolic stroke
- Stroke unit care, physiotherapy, speech therapy, occupational therapy

Prevention

- Blood pressure control
- Anticoagulation for AF

- Smoking cessation
- Statin therapy
- Antiplatelet therapy after ischemic stroke
- Lifestyle modification

When to See a Doctor

■ EMERGENCY — Call ambulance immediately. 'Time is Brain': every minute 1.9 million neurons die.

Migraine G43

Overview

A neurological disorder characterized by recurrent moderate-to-severe headaches, often on one side of the head, associated with nausea, vomiting, and sensitivity to light and sound. Affects 15% of the population.

Symptoms & Clinical Features

• Throbbing or pulsating unilateral headache	• Aura: visual disturbances, tingling, speech difficulty (in migraine with aura)
• Nausea and vomiting	• Prodrome: mood changes, neck stiffness, yawning (hours before headache)
• Photophobia (sensitivity to light)	• Postdrome: fatigue, difficulty concentrating
• Phonophobia (sensitivity to sound)	

Causes & Pathophysiology

Complex neurovascular mechanism involving trigeminovascular system activation, cortical spreading depression (in aura), and release of CGRP (calcitonin gene-related peptide).

Risk Factors

- Female sex (3:1 ratio)
- Hormonal changes (menstruation, pregnancy, menopause)
- Family history
- Sleep disruption
- Triggers: certain foods (red wine, cheese), caffeine, stress, bright lights
- Anxiety and depression

Diagnosis

Clinical diagnosis based on ICHD-3 criteria. MRI/CT to rule out secondary causes. Headache diary for trigger identification.

Treatment & Management

- Acute: NSAIDs, Triptans (Sumatriptan, Rizatriptan), CGRP antagonists (Rimegepant, Ubrogepant), antiemetics
- Preventive: Beta-blockers (Propranolol), Topiramate, Amitriptyline, CGRP monoclonal antibodies (Erenumab)
- Non-pharmacological: trigger avoidance, sleep hygiene, biofeedback, acupuncture

Prevention

- Identify and avoid personal triggers

- Maintain regular sleep schedule
- Regular meals and hydration
- Stress management
- Prophylactic medication if > 4 attacks/month

When to See a Doctor

■ Seek urgent care for 'worst headache of life', headache with fever and neck stiffness, headache after head injury, or new-onset headache over age 50.

Epilepsy G40

Overview

A neurological disorder characterized by recurrent, unprovoked seizures due to abnormal electrical activity in the brain. Affects approximately 50 million people worldwide.

Symptoms & Clinical Features

- Focal seizures: altered consciousness, automatisms, focal motor activity
- Myoclonic jerks
- Generalized tonic-clonic (grand mal): stiffening followed by jerking movements, loss of consciousness, incontinence
- Postictal confusion, headache, and fatigue
- Absence seizures: brief staring spells (especially in children)
- Aura before seizure (rising epigastric sensation, déjà vu, visual phenomena)

Causes & Pathophysiology

Idiopathic (genetic), structural (brain tumor, stroke, traumatic brain injury, cortical malformations), metabolic (hypoglycemia, electrolyte abnormalities), infectious (meningitis, encephalitis).

Risk Factors

- Family history
- Brain injuries or infections
- Stroke or vascular brain disease
- Neurodevelopmental disorders
- Age (peaks in childhood and elderly)

Diagnosis

EEG (electroencephalogram) to detect abnormal electrical activity, MRI brain, blood tests. Diagnosis requires 2 unprovoked seizures > 24 hours apart, or 1 seizure with high recurrence risk.

Treatment & Management

- Sodium Valproate, Levetiracetam, Lamotrigine, Carbamazepine, Phenytoin (drug choice depends on seizure type)
- Drug-resistant epilepsy: epilepsy surgery, vagus nerve stimulation, ketogenic diet
- Status epilepticus emergency: IV Lorazepam, Phenytoin, then general anesthesia if refractory

Prevention

- No specific primary prevention
- Medication compliance to prevent breakthrough seizures

- Avoid triggers: sleep deprivation, alcohol, photosensitivity
- Seizure first-aid education

When to See a Doctor

■ Emergency: seizure lasting > 5 minutes, second seizure without recovery, seizure in pregnant woman, seizure after head injury, first-ever seizure.

Parkinson's Disease G20

Overview

A progressive neurodegenerative disorder affecting dopaminergic neurons in the substantia nigra. Second most common neurodegenerative disease after Alzheimer's. Average onset age is 60.

Symptoms & Clinical Features

- Resting tremor (pill-rolling tremor of hands)
- Rigidity (cogwheel rigidity)
- Bradykinesia (slowness of movement)
- Postural instability and falls
- Masked facies (reduced facial expression)
- Micrographia (small handwriting)
- Shuffling gait with reduced arm swing
- Soft, monotone voice
- Depression and anxiety
- Cognitive decline in advanced stages

Causes & Pathophysiology

Loss of dopamine-producing neurons in the substantia nigra. Lewy bodies (alpha-synuclein aggregates) are the pathological hallmark. Combination of genetic and environmental factors.

Risk Factors

- Age > 60
- Male sex
- Family history
- Head injuries
- Pesticide exposure
- Rural living (well water exposure)
- LRRK2, PINK1, PARKIN gene mutations

Diagnosis

Clinical diagnosis based on UK Brain Bank Criteria (bradykinesia + 1 of rigidity/tremor/postural instability). DaTSCAN SPECT to confirm dopamine depletion. MRI to exclude other causes.

Treatment & Management

- Levodopa/Carbidopa (gold standard for motor symptoms)
- Dopamine agonists: Pramipexole, Ropinirole
- MAO-B inhibitors: Selegiline, Rasagiline
- COMT inhibitors: Entacapone
- Deep Brain Stimulation (DBS) for refractory motor symptoms
- Physiotherapy, speech therapy, occupational therapy

Prevention

- No proven prevention

- Regular exercise may be neuroprotective
- Caffeine and urate associated with reduced risk (research ongoing)

When to See a Doctor

■ **See neurologist if experiencing tremor at rest, balance problems, or significant slowness of movement.**

Gastroenterology

Gastroesophageal Reflux Disease (GERD) K21

Overview

A chronic digestive disease where stomach acid or bile flows back into the esophagus, irritating its lining. Affects approximately 20% of adults in developed countries.

Symptoms & Clinical Features

- Heartburn (burning sensation in chest)
- Regurgitation of acid or food
- Difficulty swallowing (dysphagia)
- Chronic cough
- Hoarseness or sore throat
- Sensation of lump in throat (globus)
- Chest pain mimicking cardiac disease
- Worsening after meals or when lying down

Causes & Pathophysiology

Dysfunction of lower esophageal sphincter (LES) allowing reflux. Associated with hiatal hernia, obesity, pregnancy, delayed gastric emptying.

Risk Factors

- Obesity
- Hiatal hernia
- Pregnancy
- Smoking
- Certain foods (fatty foods, chocolate, caffeine, alcohol, spicy foods)
- Certain medications (aspirin, NSAIDs, calcium channel blockers)

Diagnosis

Clinical diagnosis. Upper endoscopy (esophagogastroduodenoscopy) for complications, pH monitoring (gold standard for diagnosis), esophageal manometry.

Treatment & Management

- Proton pump inhibitors (PPIs): Omeprazole, Pantoprazole (first-line)
- H2 receptor antagonists: Ranitidine, Famotidine
- Antacids for immediate relief
- Lifestyle modifications (see prevention)
- Nissen fundoplication (surgical option for refractory GERD or large hiatal hernia)

Prevention

- Maintain healthy weight
- Avoid trigger foods and alcohol
- Eat smaller meals
- Avoid lying down within 3 hours of eating
- Elevate head of bed by 15-20cm
- Quit smoking

When to See a Doctor

■ **Seek care for dysphagia, unintentional weight loss, vomiting blood, or black tarry stools.**

Peptic Ulcer Disease K27

Overview

Open sores that develop on the inner lining of the stomach (gastric ulcers) or the upper portion of the small intestine (duodenal ulcers). H. pylori infection and NSAID use are the primary causes.

Symptoms & Clinical Features

- Burning or gnawing epigastric pain
- Vomiting blood (hematemesis)
- Pain that may improve or worsen with eating
- Unexplained weight loss
- Nausea and vomiting
- Bloating and belching
- Dark or tarry stools (melena) indicating bleeding
- Feeling of fullness after small meals

Causes & Pathophysiology

Helicobacter pylori (H. pylori) infection (70% of gastric, 90% of duodenal ulcers), chronic NSAID use, rarely Zollinger-Ellison syndrome.

Risk Factors

- H. pylori infection
- Regular NSAID use
- Smoking
- Excessive alcohol
- Stress (physiological stress: burns, ICU)
- Family history

Diagnosis

Upper endoscopy with biopsy (gold standard), urea breath test or stool antigen test for H. pylori, serology.

Treatment & Management

- H. pylori eradication: Triple therapy (PPI + Amoxicillin + Clarithromycin x 14 days)
- Discontinue NSAIDs if possible
- PPI for 4-8 weeks for healing
- Surgery for complications (perforation, obstruction, refractory bleeding)

Prevention

- Avoid H. pylori transmission (hygiene)
- Use NSAIDs cautiously with co-prescribed PPI
- Avoid smoking and excessive alcohol

When to See a Doctor

■ **Emergency for severe abdominal pain (perforation), blood in stools, or vomiting blood.**

Irritable Bowel Syndrome (IBS) K58

Overview

A functional gastrointestinal disorder characterized by a group of symptoms including abdominal pain and changes in bowel habits without any identifiable structural or biochemical explanation. Affects 10-15% of the world population.

Symptoms & Clinical Features

- Recurring abdominal pain or cramping
- Bloating and gas
- Diarrhea (IBS-D), constipation (IBS-C), or alternating (IBS-M)
- Mucus in stools
- Relief of pain with bowel movements
- Feeling of incomplete evacuation
- Symptoms worsen with stress and food

Causes & Pathophysiology

Multifactorial: altered gut motility, visceral hypersensitivity, altered gut-brain axis, dysbiosis of gut microbiome, post-infectious IBS (following gastroenteritis).

Risk Factors

- Female sex
- Age < 50
- Anxiety and depression
- Family history
- Prior GI infections
- Food sensitivities (lactose, gluten, FODMAPs)

Diagnosis

Rome IV criteria: recurrent abdominal pain >= 1 day/week for >= 3 months associated with 2+ of: related to defecation, change in stool frequency, change in stool form. Exclude IBD, celiac disease, colon cancer.

Treatment & Management

- Low-FODMAP diet
- Antispasmodics: Mebeverine, Hyoscine
- Loperamide for IBS-D
- Osmotic laxatives for IBS-C: Polyethylene glycol
- Rifaximin antibiotic for IBS-D
- Tricyclic antidepressants or SSRIs for pain modulation
- Psychological therapies: CBT, gut-directed hypnotherapy

Prevention

- Stress management
- Regular exercise
- Dietary modifications
- Regular eating habits

When to See a Doctor

■ Red flags requiring urgent evaluation: rectal bleeding, unexplained weight loss, onset after 50, nocturnal symptoms, family history of colon cancer or IBD.

Liver Cirrhosis K74

Overview

Late stage of scarring (fibrosis) of the liver caused by many forms of liver diseases and conditions. The liver slowly deteriorates and is unable to function normally. One of the leading causes of liver-related mortality.

Symptoms & Clinical Features

- Fatigue and weakness
- Easy bruising and bleeding
- Jaundice (yellowing of skin/eyes)
- Ascites (abdominal swelling from fluid)
- Spider angiomas on skin
- Palmar erythema
- Gynecomastia in men
- Hepatic encephalopathy (confusion, flapping tremor)
- Esophageal varices with risk of hemorrhage
- Pruritus (itching)

Causes & Pathophysiology

Chronic hepatitis B and C, alcoholic liver disease, non-alcoholic steatohepatitis (NASH), autoimmune hepatitis, primary biliary cholangitis, hemochromatosis, Wilson's disease.

Risk Factors

- Chronic alcohol use
- Hepatitis B or C infection
- Obesity and metabolic syndrome
- Family history of liver disease
- IV drug use

Diagnosis

Liver function tests, liver biopsy (gold standard), fibroscan (elastography), MELD score for severity, AFP for hepatocellular carcinoma surveillance, upper endoscopy for varices.

Treatment & Management

- Treat underlying cause
- Alcohol abstinence
- Antiviral therapy for hepatitis B/C
- Sodium restriction and diuretics for ascites
- Beta-blockers (Propranolol) for variceal prevention
- Lactulose and Rifaximin for hepatic encephalopathy
- Liver transplantation for end-stage disease

Prevention

- Hepatitis B vaccination
- Avoid excessive alcohol
- Safe sex practices
- Healthy weight maintenance

When to See a Doctor

■ **Seek emergency care for hematemesis from varices, severe confusion, or sudden abdominal pain.**

Endocrinology

Hypothyroidism E03

Overview

A condition in which the thyroid gland doesn't produce enough thyroid hormone. Most common endocrine disorder after diabetes. Hashimoto's thyroiditis is the most common cause in iodine-sufficient countries.

Symptoms & Clinical Features

- Fatigue and sluggishness
- Increased cold sensitivity
- Constipation
- Pale, dry skin
- Brittle nails
- Hair thinning or loss
- Puffy face
- Weight gain despite reduced appetite
- Slow heart rate (bradycardia)
- Depression and cognitive slowing
- Muscle weakness and aches
- Heavy or irregular menstrual periods

Causes & Pathophysiology

Hashimoto's thyroiditis (autoimmune), thyroid surgery, radioiodine therapy, medications (lithium, amiodarone), iodine deficiency, congenital hypothyroidism.

Risk Factors

- Female sex
- Age > 60
- Family history of thyroid disease
- Autoimmune conditions (Type 1 diabetes, lupus, rheumatoid arthritis)
- Pregnancy or postpartum period
- Prior thyroid surgery

Diagnosis

TSH (elevated), free T4 (low), thyroid antibodies (anti-TPO in Hashimoto's), thyroid ultrasound.

Treatment & Management

- Levothyroxine (synthetic T4) - daily oral replacement therapy
- Dose adjusted to normalize TSH
- Regular TSH monitoring every 6-12 months once stable

Prevention

- Adequate iodine intake
- No specific prevention for autoimmune causes

When to See a Doctor

■ Seek care for myxedema coma (severe hypothyroidism): hypothermia, extreme fatigue, low blood pressure, altered consciousness — medical emergency.

Hyperthyroidism E05

Overview

A condition in which the thyroid gland produces too much thyroid hormone. Graves' disease is the most common cause in developed countries. Leads to an accelerated metabolism.

Symptoms & Clinical Features

- Rapid or irregular heartbeat
- Unintentional weight loss
- Increased appetite
- Anxiety, nervousness, and irritability
- Tremor (fine trembling of hands)
- Sweating and heat intolerance
- Frequent bowel movements or diarrhea
- Enlarged thyroid gland (goiter)
- Fatigue and muscle weakness
- Exophthalmos (bulging eyes) in Graves' disease
- Menstrual irregularities

Causes & Pathophysiology

Graves' disease (autoimmune, most common), toxic multinodular goiter, toxic thyroid adenoma, thyroiditis, excessive iodine intake or thyroid hormone ingestion.

Risk Factors

- Female sex
- Family history
- Personal or family history of autoimmune disease
- Smoking (especially for Graves' ophthalmopathy)
- Stress
- Pregnancy

Diagnosis

TSH (suppressed), free T4 and T3 (elevated), thyroid antibodies (TSI/TRAb for Graves'), radioactive iodine uptake scan, thyroid ultrasound.

Treatment & Management

- Anti-thyroid drugs: Methimazole (Carbimazole), Propylthiouracil (PTU)
- Radioactive iodine (RAI) ablation
- Thyroidectomy
- Beta-blockers for symptom control (Propranolol)
- Glucocorticoids for Graves' ophthalmopathy

Prevention

- No specific prevention
- Avoid excess iodine supplementation if at risk

When to See a Doctor

■ **Emergency for thyroid storm: high fever, rapid heart rate, agitation, vomiting — life-threatening.**

Polycystic Ovary Syndrome (PCOS) E28.2

Overview

A hormonal disorder common among women of reproductive age. Women with PCOS may have infrequent or prolonged menstrual periods or excess male hormone (androgen) levels. The ovaries may develop numerous small collections of fluid (follicles).

Symptoms & Clinical Features

- Irregular or absent periods
- Darkening of skin in body creases
- Excess androgen: hirsutism, acne, male-pattern baldness
- Skin tags
- Polycystic ovaries on ultrasound
- Depression and anxiety
- Weight gain and difficulty losing weight
- Pelvic pain
- Infertility

Causes & Pathophysiology

Exact cause unknown. Involves insulin resistance leading to excess insulin promoting androgen production, chronic low-grade inflammation, and genetic factors.

Risk Factors

- Family history of PCOS or diabetes
- Obesity
- Insulin resistance
- Low-grade inflammation

Diagnosis

Rotterdam criteria: 2 of 3 features: oligo/anovulation, clinical/biochemical hyperandrogenism, polycystic ovaries on ultrasound. Exclude other causes of hyperandrogenism.

Treatment & Management

- Lifestyle modification: weight loss, exercise (first-line)
- Combined oral contraceptives for menstrual regulation and androgen excess
- Metformin for insulin resistance and metabolic features
- Clomiphene, Letrozole for ovulation induction in infertility
- Spironolactone for hirsutism
- Psychological support for mood disorders

Prevention

- Healthy weight maintenance
- Regular exercise
- Balanced diet low in refined carbohydrates

When to See a Doctor

■ **See gynecologist/endocrinologist for irregular periods, excessive hair growth, or difficulty conceiving.**

Infectious Diseases

Malaria B54

Overview

A life-threatening disease caused by Plasmodium parasites transmitted through the bites of infected female Anopheles mosquitoes. Approximately 240 million cases and 600,000 deaths occur annually, primarily in sub-Saharan Africa.

Symptoms & Clinical Features

• Cyclic fever and chills (every 48-72 hours depending on species)	• Fatigue
• Profuse sweating	• Jaundice
• Headache	• Dark urine (blackwater fever in severe P. falciparum)
• Nausea and vomiting	• Anemia
• Muscle pain	• Cerebral malaria: seizures, coma (severe)

Causes & Pathophysiology

Plasmodium falciparum (most deadly), P. vivax, P. malariae, P. ovale, P. knowlesi. Transmitted by Anopheles mosquito bite.

Risk Factors

- Travel to or residence in endemic area
- Lack of chemoprophylaxis
- Children under 5 in endemic areas
- Pregnant women
- Non-immune travellers
- Immunocompromised individuals

Diagnosis

Peripheral blood smear (thick and thin films) — gold standard, Rapid diagnostic tests (RDTs) for HRP-2 antigen, PCR for species identification and low parasitemia.

Treatment & Management

- Uncomplicated P. falciparum: Artemisinin-based combination therapy (ACT) — Artemether-Lumefantrine
- Severe malaria: IV Artesunate (preferred over Quinine)
- P. vivax/P. ovale: Chloroquine + Primaquine (to eliminate hypnozoites)
- Supportive care: IV fluids, antipyretics, blood transfusion if needed

Prevention

- Insecticide-treated bed nets (ITNs)
- Indoor residual spraying
- Chemoprophylaxis for travellers (Atovaquone-Proguanil, Mefloquine, Doxycycline)

- Mosquito repellents (DEET)
- RTS,S/AS01 malaria vaccine (approved 2021)

When to See a Doctor

■ Seek immediate care for high fever after travel to endemic region. Any fever in returning traveller from endemic area is malaria until proven otherwise.

Dengue Fever A90

Overview

A mosquito-borne tropical disease caused by dengue viruses (DENV 1-4) transmitted by Aedes mosquitoes. Affects 390 million people annually. Endemic in over 100 countries.

Symptoms & Clinical Features

• High fever (40 C) with sudden onset	• Nausea and vomiting
• Severe headache (retro-orbital pain)	• Skin rash appearing 3-5 days after fever
• Pain behind the eyes	• Mild bleeding (nosebleed, gum bleeding)
• Severe muscle and joint pain ('breakbone fever')	• Warning signs of severe dengue: abdominal pain, persistent vomiting, rapid breathing, blood in vomit/stools, fatigue, restlessness, decreased urine output

Causes & Pathophysiology

Four dengue virus serotypes (DENV 1-4) transmitted primarily by Aedes aegypti mosquito bites. Secondary infection with different serotype increases risk of severe dengue.

Risk Factors

- Living in or travelling to tropical regions
- Previous dengue infection (risk of severe dengue)
- Young age
- Pregnancy

Diagnosis

NS1 antigen (days 1-5), IgM/IgG antibodies (from day 5), RT-PCR, CBC (thrombocytopenia and leukopenia are hallmarks), LFTs.

Treatment & Management

- No specific antiviral treatment
- Supportive care: hydration, rest, paracetamol for fever (avoid NSAIDs and aspirin due to bleeding risk)
- Hospitalization and IV fluids for severe dengue
- Platelet transfusion for severe thrombocytopenia with bleeding

Prevention

- Eliminate Aedes mosquito breeding sites (stagnant water)
- Wear protective clothing
- Use mosquito repellents
- Dengvaxia vaccine (for seropositive individuals 9-45 years in endemic areas)

When to See a Doctor

■ **Seek urgent care for warning signs: severe abdominal pain, persistent vomiting, bleeding, rapid breathing, or decreased urine output.**

HIV/AIDS B24

Overview

HIV (Human Immunodeficiency Virus) attacks and weakens the immune system by destroying CD4+ T cells. AIDS (Acquired Immunodeficiency Syndrome) is the advanced stage of HIV infection. Approximately 38 million people live with HIV globally.

Symptoms & Clinical Features

- Acute HIV infection (2-4 weeks): fever, rash, sore throat, swollen lymph nodes, muscle aches, headache ('flu-like illness')
- Chronic infection: may be asymptomatic for years
- AIDS (CD4 < 200/mm3): recurrent opportunistic infections (PCP, toxoplasmosis, CMV, cryptococcal meningitis), weight loss > 10%, oral candidiasis, Kaposi's sarcoma

Causes & Pathophysiology

HIV-1 and HIV-2 retroviruses transmitted through: unprotected sexual intercourse, sharing needles, blood transfusion, mother-to-child (vertical transmission).

Risk Factors

- Unprotected sexual intercourse
- IV drug use with shared needles
- Multiple sexual partners
- Sexually transmitted infections
- Blood transfusions in endemic areas
- Healthcare worker needle-stick injuries

Diagnosis

HIV-1/2 antigen/antibody combination (4th generation ELISA) — screening test, confirmed by Western blot or HIV RNA PCR. CD4 count for immunological staging. Viral load for monitoring.

Treatment & Management

- Antiretroviral therapy (ART) — initiated in all HIV-positive individuals regardless of CD4 count
- Preferred regimen: Tenofovir + Emtricitabine + Dolutegravir (or Bictegravir)
- Pre-exposure prophylaxis (PrEP): Tenofovir/Emtricitabine for high-risk HIV-negative individuals
- Post-exposure prophylaxis (PEP): within 72 hours of exposure
- Treatment of opportunistic infections

Prevention

- Condom use
- HIV testing and treatment (Treatment as Prevention)
- PrEP for high-risk individuals
- Needle and syringe programs
- Prevention of mother-to-child transmission (PMTCT)

When to See a Doctor

■ **Get tested if at risk. Seek urgent care for signs of opportunistic infection or AIDS-defining illness.**

Dermatology

Psoriasis L40

Overview

A chronic immune-mediated inflammatory skin disease characterized by well-demarcated erythematous plaques with silvery-white scales. Affects approximately 2-3% of the world population. Associated with psoriatic arthritis and cardiometabolic comorbidities.

Symptoms & Clinical Features

• Red, raised, inflamed patches	• Thickened, pitted, or ridged nails
• Silvery-white scales (plaques) on red patches	• Joint swelling and stiffness (psoriatic arthritis in 30%)
• Dry, cracked skin that may bleed	• Common sites: elbows, knees, scalp, lower back
• Itching, burning, or soreness	• Koebner phenomenon (new lesions at sites of skin trauma)

Causes & Pathophysiology

Autoimmune: T-cell mediated inflammation causing accelerated skin cell cycle (days instead of weeks). Genetic (HLA-Cw6) combined with environmental triggers.

Risk Factors

- Family history (40% have first-degree relative with psoriasis)
- Stress
- Infections (streptococcal throat infection triggers guttate psoriasis)
- Certain medications (beta-blockers, lithium, antimalarials)
- Smoking and obesity
- Alcohol

Diagnosis

Clinical diagnosis. Skin biopsy if diagnosis uncertain. Assess for psoriatic arthritis with joint examination.

Treatment & Management

- Mild-moderate: Topical corticosteroids, Vitamin D analogues (Calcipotriol), Calcineurin inhibitors, Coal tar
- Moderate-severe: Phototherapy (UVB, PUVA)
- Systemic: Methotrexate, Cyclosporine, Acitretin
- Biologics: TNF-alpha inhibitors (Adalimumab, Etanercept), IL-17 inhibitors (Secukinumab), IL-23 inhibitors (Guselkumab)

Prevention

- Identify and avoid triggers
- Moisturize regularly
- Stress management

- Limit alcohol and smoking

When to See a Doctor

■ **See dermatologist for widespread disease, joint pain, or inadequate response to over-the-counter treatments.**

Atopic Dermatitis (Eczema) L20

Overview

A chronic, relapsing inflammatory skin condition characterized by intense itching and skin barrier dysfunction. Most common in children (affects 15-20% of children) but can persist into adulthood. Part of the 'atopic triad' with asthma and allergic rhinitis.

Symptoms & Clinical Features

• Intense itching (pruritus) — especially at night	• Thickened, cracked, or scaly skin (lichenification)
• Red to brownish-gray patches	• Raw, swollen skin from scratching
• Small, raised bumps that may weep fluid and crust	• Common sites: hands, feet, ankles, wrists, neck, upper chest, eyelids, inside elbows and knees

Causes & Pathophysiology

Filaggrin gene mutation causing skin barrier defect allowing allergen penetration. Immune dysregulation (Th2-dominant response). Dysbiosis with Staphylococcus aureus overgrowth.

Risk Factors

- Family history of eczema, asthma, or hay fever
- Urban living and air pollution
- Early life antibiotic use
- Living in developed countries (hygiene hypothesis)
- Female sex slightly more affected

Diagnosis

Clinical diagnosis using UK Working Party or Hanifin and Rajka criteria. IgE levels and skin prick tests for associated allergies. Swab if bacterial superinfection suspected.

Treatment & Management

- Emollients (moisturizers) — cornerstone of management; apply frequently
- Topical corticosteroids for flares
- Topical calcineurin inhibitors (Tacrolimus, Pimecrolimus)
- Dupilumab (IL-4/IL-13 receptor antagonist) for moderate-severe disease
- Antihistamines for itch
- Antibiotics/antivirals for secondary infections

Prevention

- Regular moisturization from birth in high-risk infants
- Identify and avoid triggers (certain soaps, detergents, fabrics, foods)
- Wet wrap therapy for severe flares

When to See a Doctor

■ **Seek care for signs of infection (yellow crusting, fever), severe widespread disease, or disease affecting sleep and quality of life.**

Musculoskeletal & Rheumatology

Osteoarthritis M19

Overview

The most common form of arthritis, involving the wearing away of the protective cartilage that cushions the ends of your bones. Most commonly affects joints in hands, knees, hips, and spine. Affects over 500 million people worldwide.

Symptoms & Clinical Features

- Joint pain worsening with activity
- Bone spurs (osteophytes)
- Morning stiffness lasting < 30 minutes
- Swelling around joints
- Loss of flexibility
- Bony enlargements (Heberden's and Bouchard's nodes in hands)
- Grating sensation (crepitus) on movement
- Decreased range of motion

Causes & Pathophysiology

Degeneration of articular cartilage from mechanical stress, aging, and biological factors leading to subchondral bone changes, osteophyte formation, and synovial inflammation.

Risk Factors

- Age (most common > 65 years)
- Female sex
- Obesity (especially for knee and hip OA)
- Joint injuries
- Repetitive occupational stress
- Family history
- Previous inflammatory joint disease

Diagnosis

Clinical diagnosis. X-rays show joint space narrowing, osteophytes, subchondral sclerosis, cysts. MRI for detailed assessment. Blood tests to exclude inflammatory arthritis.

Treatment & Management

- Non-pharmacological: exercise, physiotherapy, weight loss, assistive devices
- Topical NSAIDs (Diclofenac gel) for hand and knee OA
- Oral analgesics: Paracetamol, NSAIDs (with gastroprotection)
- Intra-articular corticosteroid injections
- Hyaluronic acid injections
- Total joint replacement for end-stage disease

Prevention

- Maintain healthy weight
- Regular low-impact exercise (swimming, cycling)

- Avoid joint injuries
- Strengthen muscles around joints

When to See a Doctor

■ See orthopedic surgeon or rheumatologist for severe pain limiting daily activities or if joint replacement is being considered.

Rheumatoid Arthritis M06

Overview

A chronic autoimmune inflammatory arthritis primarily affecting synovial joints. Can lead to joint destruction and systemic complications. Affects approximately 1% of the global population, predominantly women.

Symptoms & Clinical Features

- Symmetric joint inflammation (pain, swelling, warmth)
- Rheumatoid nodules
- Morning stiffness lasting > 1 hour
- Extra-articular manifestations: lung disease, vasculitis, anemia, sicca syndrome
- Small joints of hands and feet predominantly affected (MCPs, PIPs, MTPs)
- Swan-neck and boutonniere deformities in advanced disease
- Systemic features: fatigue, fever, weight loss

Causes & Pathophysiology

Autoimmune: anti-CCP and RF antibodies activate innate and adaptive immunity causing synovial inflammation, pannus formation, and joint destruction. Genetic (HLA-DRB1) and environmental (smoking) factors.

Risk Factors

- Female sex (3:1)
- Age 40-60 years
- Family history
- Smoking (strongest modifiable risk factor)
- Obesity
- Periodontitis
- Parity

Diagnosis

2010 ACR/EULAR criteria. Investigations: RF, anti-CCP (more specific), CRP/ESR, CBC, X-rays (periarticular osteopenia, erosions, joint space loss), ultrasound/MRI of joints.

Treatment & Management

- Conventional DMARDs: Methotrexate (anchor drug), Hydroxychloroquine, Sulfasalazine, Leflunomide
- Biologic DMARDs: TNF inhibitors (Adalimumab, Etanercept), IL-6 inhibitors (Tocilizumab), Rituximab
- JAK inhibitors: Tofacitinib, Baricitinib
- Corticosteroids as bridge therapy
- NSAIDs for symptom control

Prevention

- Smoking cessation is the most important modifiable preventive measure

• No other specific prevention known

When to See a Doctor

■ **Early specialist referral is crucial — within 6 weeks of symptom onset — as early treatment prevents joint damage.**

Gout M10

Overview

A form of inflammatory arthritis caused by hyperuricemia and deposition of monosodium urate crystals in joints and soft tissues. Characterized by sudden, severe attacks of pain, redness, and tenderness. Most common in the big toe (podagra).

Symptoms & Clinical Features

- Sudden, severe joint pain (often nocturnal onset)
- Tophi (urate crystal deposits) in chronic gout
- Red, warm, tender, swollen joint
- Limited range of motion during attack
- Classic site: first metatarsophalangeal joint (big toe)
- Kidney stones (urate nephrolithiasis)
- Also affects ankle, knee, wrist, elbow

Causes & Pathophysiology

Hyperuricemia (serum uric acid > 6.8 mg/dL) leads to urate crystal precipitation in joints triggering neutrophil-mediated acute inflammation. Excess uric acid from purine metabolism or reduced renal excretion.

Risk Factors

- Male sex
- Age
- Diet high in purines (red meat, organ meats, seafood)
- Alcohol (especially beer)
- Diuretic use
- Hypertension and kidney disease
- Metabolic syndrome
- Obesity

Diagnosis

Joint aspiration with identification of negatively birefringent needle-shaped monosodium urate crystals under polarized microscopy (gold standard). Serum uric acid, X-ray (erosions with overhanging edge in chronic gout), dual-energy CT.

Treatment & Management

- Acute attack: NSAIDs (Indomethacin), Colchicine, or systemic corticosteroids
- Urate-lowering therapy (ULT): Allopurinol (first-line), Febuxostat
- Target serum urate < 6 mg/dL (< 5 mg/dL in tophaceous gout)
- Pegloticase for refractory gout
- Prophylactic Colchicine or NSAID when starting ULT

Prevention

- Dietary modification: avoid red meat, organ meats, shellfish, beer, fructose
- Adequate hydration
- Maintain healthy weight
- Avoid thiazide diuretics if possible

When to See a Doctor

■ **See rheumatologist for recurrent attacks, tophi, or renal involvement.**

Nephrology (Kidney Diseases)

Chronic Kidney Disease (CKD) N18

Overview

A long-term condition where the kidneys do not work effectively. Defined by reduced GFR (< 60 mL/min/1.73m2) or markers of kidney damage for > 3 months. Affects approximately 10% of the global population.

Symptoms & Clinical Features

- Often asymptomatic in early stages
- Swelling of feet and ankles
- Fatigue and weakness
- Shortness of breath (fluid overload or anemia)
- Loss of appetite and nausea
- Hypertension
- Sleep problems
- Pruritus (itching)
- Changes in urination frequency (polyuria or oliguria)
- Uremic symptoms in advanced CKD: confusion, pericarditis, uremic frost
- Muscle cramps

Causes & Pathophysiology

Diabetes mellitus (leading cause, ~40%), hypertension (~30%), glomerulonephritis, polycystic kidney disease, recurrent UTIs, obstructive uropathy, nephrotoxic medications.

Risk Factors

- Diabetes and hypertension
- Family history of CKD
- Age > 60
- Obesity
- Smoking
- Recurrent kidney infections
- NSAIDs overuse

Diagnosis

eGFR from serum creatinine/cystatin C, urine albumin:creatinine ratio (ACR), urinalysis, renal ultrasound, kidney biopsy in selected cases. KDIGO staging (G1-G5 by GFR, A1-A3 by albuminuria).

Treatment & Management

- Treat underlying cause (diabetes: tight glycemic control; HTN: target BP < 130/80 mmHg)
- RAAS blockade: ACE inhibitors or ARBs
- SGLT-2 inhibitors: Dapagliflozin, Canagliflozin (renoprotective)
- Dietary modifications: protein and sodium restriction, potassium and phosphate control
- Management of anemia (ESAs, iron supplementation)
- Management of CKD-MBD: Phosphate binders, Vitamin D analogues
- Renal replacement therapy: Hemodialysis, Peritoneal dialysis, Kidney transplantation (GFR < 15)

Prevention

- Control diabetes and hypertension
- Avoid nephrotoxic drugs
- Adequate hydration
- Annual kidney function tests in high-risk individuals

When to See a Doctor

■ Seek care for significantly decreased urination, severe edema, shortness of breath, or confusion in known CKD patient.

Urinary Tract Infection (UTI) N39.0

Overview

An infection in any part of the urinary system — kidneys, ureters, bladder, and urethra. Most commonly involves the bladder (cystitis) and urethra (urethritis). Extremely common, especially in women (50% have at least one UTI in their lifetime).

Symptoms & Clinical Features

• Burning sensation during urination (dysuria)	• Lower abdominal discomfort
• Frequent urge to urinate	• Fever and chills (if kidney involvement: pyelonephritis)
• Cloudy or strong-smelling urine	• Flank pain (pyelonephritis)
• Pink, red, or cola-colored urine (hematuria)	• Nausea and vomiting (pyelonephritis)
• Pelvic pain (in women)	• Confusion in elderly

Causes & Pathophysiology

Escherichia coli causes 80% of uncomplicated UTIs. Other pathogens: Klebsiella, Staphylococcus saprophyticus, Enterococcus, Proteus mirabilis.

Risk Factors

- Female sex (shorter urethra)
- Sexual activity
- Menopause (reduced estrogen)
- Urinary catheterization
- Urinary tract abnormalities
- Immunosuppression
- Pregnancy
- Diabetes

Diagnosis

Urinalysis (pyuria, bacteriuria, nitrites), urine culture and sensitivity (gold standard), renal ultrasound/CT for complicated UTI or pyelonephritis.

Treatment & Management

- Uncomplicated cystitis: Nitrofurantoin, Trimethoprim-Sulfamethoxazole, or Fosfomycin (3-5 days)
- Pyelonephritis: Fluoroquinolones (Ciprofloxacin) or broad-spectrum cephalosporins (7-14 days)
- Recurrent UTI prophylaxis: daily low-dose antibiotic or post-coital antibiotic

Prevention

- Drink adequate water
- Urinate after sexual intercourse
- Wipe front to back
- Avoid potentially irritating feminine products
- Cranberry products (limited evidence)
- Topical estrogen for postmenopausal women

When to See a Doctor

■ **Seek care for high fever, severe flank pain, vomiting, symptoms in men (always evaluate), pregnant women, or failure to improve with antibiotics.**

Psychiatry & Mental Health

Major Depressive Disorder (MDD) F33

Overview

A common and serious mood disorder characterized by persistent feelings of sadness and loss of interest. Leading cause of disability worldwide. Affects approximately 280 million people globally.

Symptoms & Clinical Features

- Persistent sad, empty, or hopeless mood most of the day, nearly every day
- Fatigue or loss of energy
- Markedly diminished interest or pleasure in most activities (anhedonia)
- Feelings of worthlessness or excessive guilt
- Significant unintentional weight change or appetite change
- Difficulty thinking, concentrating, or making decisions
- Insomnia or hypersomnia
- Recurrent thoughts of death or suicide
- Psychomotor agitation or retardation

Causes & Pathophysiology

Biopsychosocial model: neurobiological (monoamine deficiency, HPA axis dysregulation), psychological (negative cognitive schemas, trauma), and social (stressors, isolation) factors interact.

Risk Factors

- Female sex (twice as common)
- Personal or family history of depression
- Major life stressors or trauma
- Chronic medical illness
- Substance use disorders
- Loneliness and social isolation
- Certain medications (beta-blockers, steroids)

Diagnosis

DSM-5 criteria: >=5 symptoms for >=2 weeks, causing significant distress/impairment, including depressed mood or anhedonia. PHQ-9 and Hamilton Depression Rating Scale for severity. Exclude medical causes (thyroid, anemia).

Treatment & Management

- Antidepressants: SSRIs (Sertraline, Escitalopram — first-line), SNRIs (Venlafaxine), TCAs, MAOIs
- Psychotherapy: Cognitive Behavioral Therapy (CBT), Interpersonal Therapy (IPT)
- Combined pharmacotherapy and psychotherapy (most effective)
- ECT for severe, treatment-resistant, or psychotic depression
- Transcranial magnetic stimulation (TMS)
- Hospitalization if suicidal risk

Prevention

- Stress management and resilience building
- Social support systems
- Regular physical activity
- Adequate sleep
- Early intervention for at-risk individuals

When to See a Doctor

■ Seek immediate care if experiencing suicidal thoughts or self-harm. Call crisis helpline if in immediate danger.

Generalized Anxiety Disorder (GAD) F41.1

Overview

A mental health condition characterized by persistent and excessive worry about various aspects of everyday life, difficult to control. Affects approximately 3.6% of the global population.

Symptoms & Clinical Features

- Excessive, uncontrollable worry about multiple life areas
- Restlessness or feeling on edge
- Easily fatigued
- Difficulty concentrating or mind going blank
- Irritability
- Muscle tension
- Sleep disturbance (difficulty falling or staying asleep)
- Somatic symptoms: headaches, GI complaints, sweating
- Avoidance behaviors

Causes & Pathophysiology

Multifactorial: hyperactivity of the amygdala, dysregulation of the HPA axis and serotonergic/GABAergic systems, genetic predisposition, and environmental stressors.

Risk Factors

- Female sex (twice as common)
- Stressful life events
- Family history of anxiety
- Childhood adversity or trauma
- Chronic medical illness
- Substance use

Diagnosis

DSM-5 criteria: excessive anxiety and worry occurring more days than not for >= 6 months. GAD-7 scale for screening and severity. Exclude medical causes (hyperthyroidism, cardiac arrhythmia).

Treatment & Management

- SSRIs (Sertraline, Escitalopram) and SNRIs (Venlafaxine, Duloxetine) — first-line
- Buspirone (non-benzodiazepine anxiolytic)
- Short-term Benzodiazepines (risk of dependence)
- CBT (most evidence-based psychological therapy)

- Mindfulness-based interventions
- Relaxation techniques, exercise

Prevention

- Stress reduction techniques
- Regular physical activity
- Good sleep hygiene
- Limit caffeine and alcohol
- Social connectedness

When to See a Doctor

■ **See GP or mental health professional if anxiety is significantly impairing daily function, relationships, or work.**

Oncology (Cancer Medicine)

Breast Cancer C50

Overview

The most common cancer in women worldwide, accounting for approximately 12% of all cancer diagnoses globally. Can also occur in men. Most cases are adenocarcinomas arising from ductal or lobular epithelium.

Symptoms & Clinical Features

- Painless breast lump (most common presentation)
- Nipple discharge (bloody or clear)
- Change in breast size or shape
- Swollen lymph nodes in axilla
- Skin changes: dimpling, peau d'orange appearance
- Breast pain (less common)
- Nipple retraction or inversion
- Redness, flaking, or thickening of nipple skin (Paget's disease)

Causes & Pathophysiology

Multifactorial: hormonal (estrogen exposure), genetic (BRCA1/BRCA2 mutations in 5-10%), environmental, and lifestyle factors causing uncontrolled proliferation of breast epithelial cells.

Risk Factors

- Female sex and increasing age
- BRCA1 and BRCA2 gene mutations
- Family history of breast or ovarian cancer
- Dense breast tissue
- Hormone replacement therapy (HRT)
- Nulliparity or first pregnancy after age 30
- Early menarche/late menopause
- Obesity (postmenopausal)
- Alcohol consumption
- Prior chest radiation

Diagnosis

Mammography (screening), ultrasound, MRI breast, core needle biopsy (gold standard for histological diagnosis), ER/PR/HER2 receptor status, staging CT/bone scan/PET.

Treatment & Management

- Surgery: breast-conserving surgery (lumpectomy) or mastectomy with sentinel/axillary lymph node assessment
- Radiotherapy after breast-conserving surgery
- Hormonal therapy for ER/PR positive: Tamoxifen (premenopausal), Aromatase inhibitors (postmenopausal)
- HER2-targeted therapy: Trastuzumab, Pertuzumab
- Chemotherapy: Anthracyclines, Taxanes

- CDK4/6 inhibitors for metastatic ER+ disease: Palbociclib

Prevention

- Breast self-examination awareness
- Regular mammographic screening (from age 40-50 per guidelines)
- Preventive surgery or chemoprevention for BRCA carriers
- Limit alcohol, maintain healthy weight, exercise
- Breastfeeding has protective effect

When to See a Doctor

■ **Immediately for any new breast lump, skin changes, nipple discharge, or swollen lymph nodes.**

Colorectal Cancer C18-C20

Overview

Cancer arising from the colon or rectum, collectively called colorectal cancer (CRC). Third most common cancer worldwide and second most common cause of cancer death. Most cases develop from adenomatous polyps.

Symptoms & Clinical Features

- Change in bowel habits (diarrhea, constipation, narrowing of stool)
- Unexplained iron-deficiency anemia
- Rectal bleeding or blood in stool
- Unexplained weight loss
- Persistent abdominal discomfort, cramps, or pain
- Fatigue
- Sensation of incomplete bowel emptying
- Emergency presentation with obstruction or perforation

Causes & Pathophysiology

Progression from adenomatous polyp to invasive carcinoma (adenoma-carcinoma sequence). Sporadic (80%), hereditary (Lynch syndrome, FAP) (20%). Involves mutations in APC, KRAS, TP53.

Risk Factors

- Age > 50 (90% of cases)
- Family history of CRC or polyps
- Hereditary syndromes (Lynch syndrome, FAP)
- Inflammatory bowel disease (UC, Crohn's)
- Diet high in red/processed meat
- Obesity and physical inactivity
- Smoking and alcohol
- Type 2 diabetes

Diagnosis

Colonoscopy with biopsy (gold standard), CT colonography, flexible sigmoidoscopy, FIT (fecal immunochemical test) for screening, CEA (tumor marker), staging CT chest/abdomen/pelvis.

Treatment & Management

- Surgery: hemicolectomy, anterior resection, or abdominoperineal resection

- Adjuvant chemotherapy: FOLFOX, CAPOX for stage III
- Neoadjuvant chemoradiotherapy for locally advanced rectal cancer
- Targeted therapy: Bevacizumab (anti-VEGF), Cetuximab (anti-EGFR for RAS wild-type)
- Immunotherapy: Pembrolizumab for MSI-H/dMMR tumours

Prevention

- Regular colonoscopy screening from age 45-50 (or 10 years before youngest affected relative)
- High-fiber, low-red-meat diet
- Regular physical activity
- Maintain healthy weight
- Aspirin (in selected high-risk individuals)
- Polypectomy of adenomas at colonoscopy

When to See a Doctor

■ **Seek urgent care for rectal bleeding, unexplained weight loss, or change in bowel habits lasting > 3 weeks.**

Ophthalmology (Eye Diseases)

Glaucoma H40

Overview

A group of eye conditions that damage the optic nerve, often associated with elevated intraocular pressure (IOP). Leading cause of irreversible blindness worldwide. Primary open-angle glaucoma (POAG) is the most common form.

Symptoms & Clinical Features

- POAG: often asymptomatic until late stages — 'silent thief of sight'
- Tunnel vision in advanced disease
- Gradual loss of peripheral (side) vision
- Acute angle-closure glaucoma: sudden severe eye pain, headache, nausea, halos around lights, blurred vision, redness of eye

Causes & Pathophysiology

Elevated IOP (> 21 mmHg) damages optic nerve axons. In POAG: impaired drainage via trabecular meshwork. In acute angle-closure: blockage of aqueous outflow at iridocorneal angle.

Risk Factors

- IOP > 21 mmHg (ocular hypertension)
- Age > 60
- Family history of glaucoma
- African or Hispanic ancestry
- Myopia (POAG)
- Hyperopia and shallow anterior chamber (angle-closure)
- Corticosteroid use
- Diabetes
- Previous eye injury

Diagnosis

Tonometry (IOP measurement), visual field testing (perimetry), optic disc assessment (cup-disc ratio > 0.6 suspicious), OCT of retinal nerve fiber layer and optic disc, gonioscopy.

Treatment & Management

- Medical: Prostaglandin analogues (Latanoprost — first-line), Beta-blockers (Timolol), Alpha-agonists (Brimonidine), Carbonic anhydrase inhibitors (Dorzolamide)
- Laser: Selective Laser Trabeculoplasty (SLT) for POAG, Laser peripheral iridotomy for angle-closure
- Surgery: Trabeculectomy, Glaucoma drainage devices for refractory cases
- Acute angle-closure: IV Acetazolamide, topical agents, then laser iridotomy

Prevention

- Regular comprehensive eye examinations (every 1-2 years for > 40 years)
- Protective eyewear to prevent eye injuries

- Eye drops compliance in diagnosed patients

When to See a Doctor

■ Emergency for acute angle-closure: sudden eye pain, vision loss, halos, nausea. Regular check-ups for early POAG detection.

Hematology (Blood Disorders)

Iron Deficiency Anemia D50

Overview

The most common nutritional deficiency and the most prevalent cause of anemia worldwide. Affects approximately 1.6 billion people. Results from inadequate iron for hemoglobin synthesis.

Symptoms & Clinical Features

- Fatigue and weakness
- Pale skin and conjunctivae
- Shortness of breath on exertion
- Dizziness or lightheadedness
- Cold hands and feet
- Brittle nails (koilonychia — spoon-shaped nails)
- Unusual cravings for non-nutritive substances: ice, dirt, starch (pica)
- Restless leg syndrome
- Headache
- Tachycardia
- Glossitis and angular stomatitis
- Hair loss

Causes & Pathophysiology

Inadequate dietary iron intake, poor absorption (celiac disease, H. pylori, post-gastrectomy), increased requirement (pregnancy, growth), chronic blood loss (GI bleeding, heavy menstruation).

Risk Factors

- Premenopausal women
- Pregnancy
- Infants and young children
- Vegetarians and vegans
- Frequent blood donors
- GI conditions affecting absorption

Diagnosis

CBC (low Hb, microcytic hypochromic anemia, high RDW), low serum ferritin (most sensitive early indicator), low serum iron, high TIBC, low transferrin saturation. Investigate underlying cause.

Treatment & Management

- Oral iron supplements: Ferrous sulfate 200mg TDS (with Vitamin C to enhance absorption)
- Take on empty stomach if tolerated
- IV iron (Ferric carboxymaltose) for intolerance or malabsorption
- Treat underlying cause (e.g., stop NSAIDs, manage heavy periods, treat celiac disease)
- Dietary advice: red meat, legumes, leafy greens, fortified cereals

Prevention

- Dietary diversification and iron-rich foods
- Iron supplementation in pregnancy

• Food fortification programs
• Treat H. pylori in endemic areas

When to See a Doctor

■ Seek care for significant fatigue, very pale appearance, or any signs of GI bleeding (rectal bleeding, dark stools).

Deep Vein Thrombosis (DVT) I82

Overview

Formation of a blood clot (thrombus) in a deep vein, usually in the legs. Can lead to pulmonary embolism (PE) if the clot breaks off and travels to the lungs, making it potentially life-threatening.

Symptoms & Clinical Features

• Leg pain or tenderness (often in calf — Homan's sign, not reliable)	• Distended surface veins
• Swelling of affected leg	• May be asymptomatic (50% of cases)
• Redness and warmth over affected area	• Pulmonary embolism symptoms if clot embolizes: sudden breathlessness, chest pain, haemoptysis, collapse

Causes & Pathophysiology

Virchow's triad: venous stasis, endothelial injury, and hypercoagulability. Caused by prolonged immobility, trauma, surgery, malignancy, thrombophilia, pregnancy, OCP use.

Risk Factors

• Prolonged immobility (long-haul flights, bed rest)
• Recent surgery (especially hip/knee replacement)
• Malignancy
• Thrombophilia (Factor V Leiden, antiphospholipid syndrome)
• Pregnancy and postpartum
• Combined oral contraceptive pill
• Obesity
• Previous DVT/PE
• Age > 60

Diagnosis

Wells score for pre-test probability, D-dimer (highly sensitive; if normal rules out DVT in low probability), compression Doppler ultrasound (first-line imaging). CT pulmonary angiography if PE suspected.

Treatment & Management

• Anticoagulation: DOACs (Rivaroxaban, Apixaban — first-line), LMWH, Warfarin
• Duration: 3 months for provoked DVT, indefinite for unprovoked or recurrent DVT or active malignancy
• Compression stockings
• Thrombolysis for massive PE with hemodynamic instability
• IVC filter if anticoagulation contraindicated

Prevention

• Mechanical prophylaxis: compression stockings, pneumatic compression devices

- Pharmacological prophylaxis: LMWH in high-risk surgical patients
- Early ambulation after surgery
- Stay hydrated during long journeys

When to See a Doctor

■ **Emergency for sudden breathlessness, chest pain, rapid heart rate, or collapse (PE). Urgent care for new significant leg swelling and pain.**

Pediatrics (Childhood Diseases)

Acute Otitis Media (Middle Ear Infection) H66

Overview

An infection of the middle ear, extremely common in young children. By 3 years of age, 80% of children have had at least one episode. Leading reason for antibiotic prescriptions in pediatric practice.

Symptoms & Clinical Features

- Ear pain (otalgia) — infants may pull or tug at ears
- Hearing difficulties
- Fever
- Loss of appetite
- Irritability and crying (especially in infants)
- Ear drainage (otorrhea) — indicates eardrum rupture
- Difficulty sleeping
- Vomiting and diarrhea in young children

Causes & Pathophysiology

Bacterial: Streptococcus pneumoniae, Haemophilus influenzae, Moraxella catarrhalis. Viral origin in 30-40%. Often follows viral upper respiratory tract infection causing Eustachian tube dysfunction.

Risk Factors

- Age 6-24 months (peak incidence)
- Daycare attendance
- Bottle feeding (vs. breastfeeding)
- Smoking in household
- Cleft palate
- Down syndrome
- Winter season

Diagnosis

Pneumatic otoscopy showing erythema, bulging, opacity or perforation of tympanic membrane with reduced mobility. Tympanometry, audiometry for recurrent disease.

Treatment & Management

- Watchful waiting for 48-72 hours in mild-moderate cases in children > 2 years (most resolve spontaneously)
- Amoxicillin (80-90 mg/kg/day) for 5-10 days (first-line antibiotic)
- Amoxicillin-Clavulanate for treatment failure
- Paracetamol/Ibuprofen for pain and fever
- Tympanostomy tubes (grommets) for recurrent AOM (>= 3 episodes in 6 months)

Prevention

- Breastfeeding for >= 6 months
- Pneumococcal and influenza vaccination
- Avoid passive smoking

• Avoid bottle propping in supine position

When to See a Doctor

■ Seek care for ear pain with fever in children under 2 immediately. Seek urgent care for severe pain, mastoid swelling, facial weakness, or altered consciousness.

Kawasaki Disease M30.3

Overview

A febrile vasculitis of medium-sized vessels occurring predominantly in children < 5 years. Most common cause of acquired heart disease in children in developed countries due to coronary artery aneurysm formation.

Symptoms & Clinical Features

• Fever lasting >= 5 days (sine qua non)	• Changes in extremities: erythema/edema of hands and feet, periungual desquamation (late sign)
• Bilateral non-exudative conjunctival injection	• Cervical lymphadenopathy (>= 1.5 cm)
• Changes in lips and oral cavity: strawberry tongue, red cracked lips	• Irritability
• Polymorphous rash	• Coronary artery aneurysms (in untreated patients: 20-25%)

Causes & Pathophysiology

Unknown etiology. Likely triggered by an infectious agent (possibly respiratory virus) in genetically susceptible children causing aberrant immune activation and systemic vasculitis.

Risk Factors

• Asian ancestry (especially Japanese, Korean)
• Age 6 months to 5 years
• Male sex (1.5:1)
• Siblings of affected children

Diagnosis

Clinical criteria: fever >= 5 days + >= 4 of 5 features. Investigations: ESR/CRP elevated, anemia, thrombocytosis (late), elevated transaminases, echocardiography (for coronary aneurysms).

Treatment & Management

• High-dose aspirin (30-50 mg/kg/day) during acute phase
• IVIG (2g/kg single dose): reduces fever and coronary aneurysm risk from 20% to <5% if given within 10 days of fever onset
• Low-dose aspirin long-term for those with coronary involvement
• Infliximab for IVIG-resistant Kawasaki disease

Prevention

• No known prevention
• Early diagnosis and treatment critical to prevent coronary aneurysms

When to See a Doctor

■ **Urgent evaluation for any child with unexplained fever lasting > 5 days, especially with rash, red eyes, or mouth changes.**

Gynecology & Obstetrics

Endometriosis N80

Overview

A condition where tissue similar to the lining of the uterus (endometrium) grows outside the uterus — commonly on ovaries, fallopian tubes, and the lining of the pelvis. Affects approximately 10% of women of reproductive age.

Symptoms & Clinical Features

- Severe dysmenorrhea (painful periods)
- Chronic pelvic pain
- Dyspareunia (painful intercourse)
- Dyschezia (painful bowel movements, especially during menstruation)
- Dysuria (painful urination during menstruation)
- Heavy menstrual bleeding or bleeding between periods
- Infertility (40-50% of infertile women have endometriosis)
- Fatigue
- Bloating ('endo belly')
- Nausea and vomiting

Causes & Pathophysiology

Not fully understood. Theories include retrograde menstruation (Sampson's theory), coelomic metaplasia, lymphovascular spread. Immune system dysfunction allows ectopic endometrial tissue to survive and grow.

Risk Factors

- Family history
- Short menstrual cycle (< 27 days)
- Heavy/prolonged periods
- Never having given birth
- Starting periods early
- Going through menopause late
- High estrogen levels
- Low BMI

Diagnosis

Laparoscopy with biopsy (gold standard). Transvaginal ultrasound and MRI for advanced disease mapping (endometriomas, deep infiltrating endometriosis). CA-125 (elevated but not specific).

Treatment & Management

- Pain management: NSAIDs, paracetamol
- Hormonal suppression: combined OCP, progestins, GnRH agonists (Leuprorelin), levonorgestrel IUD
- Surgical: conservative laparoscopic excision or ablation of lesions
- Hysterectomy with bilateral salpingo-oophorectomy for refractory cases without fertility desire
- Assisted reproduction (IVF) for associated infertility

Prevention

- No proven prevention
- Hormonal contraception may reduce progression
- Early diagnosis and treatment to preserve fertility

When to See a Doctor

■ **See gynecologist for severe painful periods, chronic pelvic pain, or difficulty conceiving.**

Preeclampsia O14

Overview

A pregnancy complication characterized by new-onset hypertension (BP >= 140/90 mmHg) after 20 weeks of gestation with organ dysfunction. A leading cause of maternal and perinatal morbidity and mortality worldwide.

Symptoms & Clinical Features

- New-onset hypertension after 20 weeks
- Proteinuria
- Severe headache
- Visual disturbances (flashing lights, blurred vision)
- Upper abdominal or epigastric pain (right upper quadrant — hepatic involvement)
- Nausea and vomiting
- Facial swelling
- Sudden significant weight gain from edema
- HELLP syndrome: haemolysis, elevated liver enzymes, low platelets

Causes & Pathophysiology

Incomplete trophoblast invasion of spiral arteries leading to placental ischemia, systemic endothelial dysfunction, oxidative stress, and widespread vascular involvement.

Risk Factors

- First pregnancy
- Prior history of preeclampsia
- Multifetal pregnancy
- Chronic hypertension or kidney disease
- Diabetes
- Obesity
- Age < 20 or > 35
- Family history
- Autoimmune conditions

Diagnosis

BP >= 140/90 mmHg on two occasions 4 hours apart after 20 weeks, with proteinuria (>= 300 mg/24h) or end-organ dysfunction. Investigations: 24-hour urine protein, CBC, LFTs, renal function, uric acid, fetal assessment.

Treatment & Management

- Delivery is the definitive treatment
- Antihypertensives: Labetalol, Nifedipine, Methyldopa to prevent maternal stroke
- Magnesium sulfate for seizure prophylaxis (eclampsia prevention)
- Corticosteroids for fetal lung maturity if < 34 weeks

- Aspirin 75-150 mg/day from 12 weeks for women at high risk of preeclampsia

Prevention

- Low-dose aspirin from 12-16 weeks in high-risk women
- Calcium supplementation in low-intake populations
- Regular antenatal monitoring of BP and urine protein

When to See a Doctor

■ **Emergency for severe headache, visual disturbances, severe epigastric pain, or any seizure during pregnancy.**

Symptom Quick Reference Index

Use this index to quickly identify diseases associated with a given symptom. Ideal for RAG retrieval by symptom keyword.

Symptom / Feature	Associated Conditions
AIDS	HIV/AIDS
Abdominal bloating and nausea	Heart Failure
Absence seizures	Epilepsy
Acute HIV infection	HIV/AIDS
Acute angle-closure glaucoma	Glaucoma
Also affects ankle, knee, wrist, elbow	Gout
Anemia	Malaria
Anxiety, nervousness, and irritability	Hyperthyroidism
Ascites	Liver Cirrhosis
Aura	Migraine
Aura before seizure	Epilepsy
Avoidance behaviors	Generalized Anxiety Disorder (GAD)
Barrel chest	Chronic Obstructive Pulmonary Disease (COPD)
Bilateral ankle and leg edema	Heart Failure
Bilateral non-exudative conjunctival injection	Kawasaki Disease
Bloating	Endometriosis
Bloating and belching	Peptic Ulcer Disease
Bloating and gas	Irritable Bowel Syndrome (IBS)
Blurred vision	Type 2 Diabetes Mellitus, Hypertension (High Blood Pressure)
Body aches	Common Cold (Upper Respiratory Tract Infection)
Bone spurs	Osteoarthritis
Bony enlargements	Osteoarthritis
Bradykinesia	Parkinson's Disease
Breast pain	Breast Cancer
Brittle nails	Hypothyroidism, Iron Deficiency Anemia
Burning or gnawing epigastric pain	Peptic Ulcer Disease
Burning sensation during urination	Urinary Tract Infection (UTI)
Cerebral malaria	Malaria
Cervical lymphadenopathy	Kawasaki Disease

Change in bowel habits	Colorectal Cancer
Change in breast size or shape	Breast Cancer
Changes in extremities	Kawasaki Disease
Changes in lips and oral cavity	Kawasaki Disease
Changes in urination frequency	Chronic Kidney Disease (CKD)
Chest pain	Hypertension (High Blood Pressure), Pulmonary Tuberculosis
Chest pain mimicking cardiac disease	Gastroesophageal Reflux Disease (GERD)
Chest pain or discomfort	Atrial Fibrillation
Chest tightness	Asthma, Chronic Obstructive Pulmonary Disease (COPD)
Chills	Influenza (Flu)
Chronic cough	Gastroesophageal Reflux Disease (GERD)
Chronic infection	HIV/AIDS
Chronic pelvic pain	Endometriosis
Chronic productive cough	Chronic Obstructive Pulmonary Disease (COPD)
Classic site	Gout
Cloudy or strong-smelling urine	Urinary Tract Infection (UTI)
Cognitive decline in advanced stages	Parkinson's Disease
Cold hands and feet	Iron Deficiency Anemia
Cold sweats	Acute Myocardial Infarction (Heart Attack)
Common sites	Psoriasis, Atopic Dermatitis (Eczema)
Confusion	Pneumonia
Confusion in elderly	Urinary Tract Infection (UTI)
Constipation	Hypothyroidism
Coronary artery aneurysms	Kawasaki Disease
Cough	Common Cold (Upper Respiratory Tract Infection), Asthma
Coughing up blood	Pulmonary Tuberculosis
Crushing chest pain or pressure radiating to arm,	Acute Myocardial Infarction (Heart Attack)
Cyanosis	Chronic Obstructive Pulmonary Disease (COPD)
Cyclic fever and chills	Malaria
Dark or tarry stools	Peptic Ulcer Disease
Dark urine	Malaria
Darkened skin in body creases	Type 2 Diabetes Mellitus
Darkening of skin in body creases	Polycystic Ovary Syndrome (PCOS)
Decreased range of motion	Osteoarthritis

Depression and anxiety	Parkinson's Disease, Polycystic Ovary Syndrome (PCOS)
Depression and cognitive slowing	Hypothyroidism
Diarrhea	Irritable Bowel Syndrome (IBS)
Difficulty concentrating or mind going blank	Generalized Anxiety Disorder (GAD)
Difficulty sleeping	Acute Otitis Media (Middle Ear Infection)
Difficulty sleeping due to breathing problems	Asthma
Difficulty swallowing	Gastroesophageal Reflux Disease (GERD)
Difficulty thinking, concentrating, or making deci	Major Depressive Disorder (MDD)
Distended surface veins	Deep Vein Thrombosis (DVT)
Dizziness	Hypertension (High Blood Pressure)
Dizziness or lightheadedness	Atrial Fibrillation, Iron Deficiency Anemia
Dry cough	Influenza (Flu)
Dry, cracked skin that may bleed	Psoriasis
Dyschezia	Endometriosis
Dyspareunia	Endometriosis
Dyspnea	Chronic Obstructive Pulmonary Disease (COPD)
Dyspnea on exertion	Heart Failure
Dysuria	Endometriosis
Ear drainage	Acute Otitis Media (Middle Ear Infection)
Ear pain	Acute Otitis Media (Middle Ear Infection)
Easily fatigued	Generalized Anxiety Disorder (GAD)
Easy bruising and bleeding	Liver Cirrhosis
Emergency presentation with obstruction or perfora	Colorectal Cancer
Enlarged thyroid gland	Hyperthyroidism
Esophageal varices with risk of hemorrhage	Liver Cirrhosis
Excess androgen	Polycystic Ovary Syndrome (PCOS)
Excessive, uncontrollable worry about multiple lif	Generalized Anxiety Disorder (GAD)
Exophthalmos	Hyperthyroidism
Extra-articular manifestations	Rheumatoid Arthritis
FAST	Stroke (Cerebrovascular Accident)
Facial swelling	Preeclampsia

Fatigue	Type 2 Diabetes Mellitus, Pneumonia, Pulmonary Tuberculosis, Acute Myocardial Infarction (Heart Attack), Atrial Fibrillation, Malaria, Colorectal Cancer, Endometriosis
Fatigue and muscle weakness	Hyperthyroidism
Fatigue and sluggishness	Hypothyroidism
Fatigue and weakness	Influenza (Flu), Heart Failure, Liver Cirrhosis, Chronic Kidney Disease (CKD), Iron Deficiency Anemia
Fatigue during exercise	Asthma
Fatigue or loss of energy	Major Depressive Disorder (MDD)
Feeling of fullness after small meals	Peptic Ulcer Disease
Feeling of incomplete evacuation	Irritable Bowel Syndrome (IBS)
Feelings of worthlessness or excessive guilt	Major Depressive Disorder (MDD)
Fever	Pulmonary Tuberculosis, Acute Otitis Media (Middle Ear Infection)
Fever and chills	Urinary Tract Infection (UTI)
Fever lasting >= 5 days	Kawasaki Disease
Flank pain	Urinary Tract Infection (UTI)
Focal seizures	Epilepsy
Frequent bowel movements or diarrhea	Hyperthyroidism
Frequent respiratory infections	Chronic Obstructive Pulmonary Disease (COPD)
Frequent urge to urinate	Urinary Tract Infection (UTI)
Frequent urination	Type 2 Diabetes Mellitus
Generalized tonic-clonic	Epilepsy
Glossitis and angular stomatitis	Iron Deficiency Anemia
Gradual loss of peripheral	Glaucoma
Grating sensation	Osteoarthritis
Gynecomastia in men	Liver Cirrhosis
HELLP syndrome	Preeclampsia
Hair loss	Iron Deficiency Anemia
Hair thinning or loss	Hypothyroidism
Headache	Hypertension (High Blood Pressure), Malaria, Iron Deficiency Anemia
Hearing difficulties	Acute Otitis Media (Middle Ear Infection)
Heartburn	Gastroesophageal Reflux Disease (GERD)
Heavy menstrual bleeding or bleeding between perio	Endometriosis
Heavy or irregular menstrual periods	Hypothyroidism
Hepatic encephalopathy	Liver Cirrhosis

High fever	Influenza (Flu), Dengue Fever
High fever with chills	Pneumonia
Hoarseness or sore throat	Gastroesophageal Reflux Disease (GERD)
Hypertension	Chronic Kidney Disease (CKD)
Increased appetite	Hyperthyroidism
Increased cold sensitivity	Hypothyroidism
Increased thirst	Type 2 Diabetes Mellitus
Infertility	Polycystic Ovary Syndrome (PCOS), Endometriosis
Insomnia or hypersomnia	Major Depressive Disorder (MDD)
Intense itching	Atopic Dermatitis (Eczema)
Irregular or absent periods	Polycystic Ovary Syndrome (PCOS)
Irritability	Generalized Anxiety Disorder (GAD), Kawasaki Disease
Irritability and crying	Acute Otitis Media (Middle Ear Infection)
Itching, burning, or soreness	Psoriasis
Jaundice	Liver Cirrhosis, Malaria
Joint pain worsening with activity	Osteoarthritis
Joint swelling and stiffness	Psoriasis
Kidney stones	Gout
Koebner phenomenon	Psoriasis
Leg pain or tenderness	Deep Vein Thrombosis (DVT)
Lightheadedness or dizziness	Acute Myocardial Infarction (Heart Attack)
Limited range of motion during attack	Gout
Loss of appetite	Influenza (Flu), Pulmonary Tuberculosis, Acute Otitis Media (Middle Ear Infection)
Loss of appetite and nausea	Chronic Kidney Disease (CKD)
Loss of flexibility	Osteoarthritis
Loss of smell/taste	Common Cold (Upper Respiratory Tract Infection)
Low oxygen saturation	Pneumonia
Low-grade fever	Common Cold (Upper Respiratory Tract Infection)
Lower abdominal discomfort	Urinary Tract Infection (UTI)
Lymph node enlargement	Pulmonary Tuberculosis
Markedly diminished interest or pleasure in most a	Major Depressive Disorder (MDD)
Masked facies	Parkinson's Disease
May be asymptomatic	Deep Vein Thrombosis (DVT)
May be completely asymptomatic	Atrial Fibrillation

Menstrual irregularities	Hyperthyroidism
Micrographia	Parkinson's Disease
Mild bleeding	Dengue Fever
Mild fatigue	Common Cold (Upper Respiratory Tract Infection)
Mild headache	Common Cold (Upper Respiratory Tract Infection)
Morning stiffness lasting < 30 minutes	Osteoarthritis
Morning stiffness lasting > 1 hour	Rheumatoid Arthritis
Mucus in stools	Irritable Bowel Syndrome (IBS)
Muscle and body aches	Influenza (Flu)
Muscle cramps	Chronic Kidney Disease (CKD)
Muscle pain	Malaria
Muscle tension	Generalized Anxiety Disorder (GAD)
Muscle weakness and aches	Hypothyroidism
Myoclonic jerks	Epilepsy
Nausea and vomiting	Pneumonia, Acute Myocardial Infarction (Heart Attack), Migraine, Peptic Ulcer Disease, Malaria, Dengue Fever, Urinary Tract Infection (UTI), Endometriosis, Preeclampsia
New-onset hypertension after 20 weeks	Preeclampsia
Night sweats	Pulmonary Tuberculosis
Nipple discharge	Breast Cancer
Nipple retraction or inversion	Breast Cancer
Nosebleeds	Hypertension (High Blood Pressure)
Occasional vomiting/diarrhea	Influenza (Flu)
Often asymptomatic in early stages	Chronic Kidney Disease (CKD)
Orthopnea	Heart Failure
POAG	Glaucoma
Pain behind the eyes	Dengue Fever
Pain that may improve or worsen with eating	Peptic Ulcer Disease
Painless breast lump	Breast Cancer
Pale skin and conjunctivae	Iron Deficiency Anemia
Pale, dry skin	Hypothyroidism
Palmar erythema	Liver Cirrhosis
Palpitations	Acute Myocardial Infarction (Heart Attack), Atrial Fibrillation
Paroxysmal nocturnal dyspnea	Heart Failure
Pelvic pain	Polycystic Ovary Syndrome (PCOS), Urinary Tract Infection (UTI)

Persistent abdominal discomfort, cramps, or pain	Colorectal Cancer
Persistent cough lasting more than 3 weeks	Pulmonary Tuberculosis
Persistent cough or wheezing	Heart Failure
Persistent sad, empty, or hopeless mood most of th	Major Depressive Disorder (MDD)
Phonophobia	Migraine
Photophobia	Migraine
Pink, red, or cola-colored urine	Urinary Tract Infection (UTI)
Pleuritic chest pain	Pneumonia
Polycystic ovaries on ultrasound	Polycystic Ovary Syndrome (PCOS)
Polymorphous rash	Kawasaki Disease
Postdrome	Migraine
Postictal confusion, headache, and fatigue	Epilepsy
Postural instability and falls	Parkinson's Disease
Prodrome	Migraine
Productive cough with rust-colored or greenish spu	Pneumonia
Profuse sweating	Malaria
Proteinuria	Preeclampsia
Pruritus	Liver Cirrhosis, Chronic Kidney Disease (CKD)
Psychomotor agitation or retardation	Major Depressive Disorder (MDD)
Puffy face	Hypothyroidism
Pulmonary embolism symptoms if clot embolizes	Deep Vein Thrombosis (DVT)
Rapid breathing	Asthma, Pneumonia
Rapid heart rate	Pneumonia
Rapid or irregular heartbeat	Heart Failure, Hyperthyroidism
Rapid weight gain from fluid retention	Heart Failure
Raw, swollen skin from scratching	Atopic Dermatitis (Eczema)
Rectal bleeding or blood in stool	Colorectal Cancer
Recurrent thoughts of death or suicide	Major Depressive Disorder (MDD)
Recurring abdominal pain or cramping	Irritable Bowel Syndrome (IBS)
Red to brownish-gray patches	Atopic Dermatitis (Eczema)

Red, raised, inflamed patches	Psoriasis
Red, warm, tender, swollen joint	Gout
Redness and warmth over affected area	Deep Vein Thrombosis (DVT)
Redness, flaking, or thickening of nipple skin	Breast Cancer
Reduced exercise tolerance	Atrial Fibrillation
Regurgitation of acid or food	Gastroesophageal Reflux Disease (GERD)
Relief of pain with bowel movements	Irritable Bowel Syndrome (IBS)
Resting tremor	Parkinson's Disease
Restless leg syndrome	Iron Deficiency Anemia
Restlessness or feeling on edge	Generalized Anxiety Disorder (GAD)
Rheumatoid nodules	Rheumatoid Arthritis
Rigidity	Parkinson's Disease
Runny or stuffy nose	Influenza (Flu), Common Cold (Upper Respiratory Tract Infection)
Sensation of incomplete bowel emptying	Colorectal Cancer
Sensation of lump in throat	Gastroesophageal Reflux Disease (GERD)
Severe dysmenorrhea	Endometriosis
Severe headache	Influenza (Flu), Dengue Fever, Preeclampsia
Severe muscle and joint pain	Dengue Fever
Shortness of breath	Hypertension (High Blood Pressure), Asthma, Pneumonia, Acute Myocardial Infarction (Heart Attack), Atrial Fibrillation, Chronic Kidney Disease (CKD)
Shortness of breath on exertion	Iron Deficiency Anemia
Shuffling gait with reduced arm swing	Parkinson's Disease
Significant unintentional weight change or appetit	Major Depressive Disorder (MDD)
Silent MI may be asymptomatic	Acute Myocardial Infarction (Heart Attack)
Silvery-white scales	Psoriasis
Skin changes	Breast Cancer
Skin rash appearing 3-5 days after fever	Dengue Fever
Skin tags	Polycystic Ovary Syndrome (PCOS)
Sleep disturbance	Generalized Anxiety Disorder (GAD)
Sleep problems	Chronic Kidney Disease (CKD)
Slow heart rate	Hypothyroidism
Slow-healing sores or frequent infections	Type 2 Diabetes Mellitus

Small joints of hands and feet predominantly affec	Rheumatoid Arthritis
Small, raised bumps that may weep fluid and crust	Atopic Dermatitis (Eczema)
Sneezing	Common Cold (Upper Respiratory Tract Infection)
Soft, monotone voice	Parkinson's Disease
Somatic symptoms	Generalized Anxiety Disorder (GAD)
Sore throat	Influenza (Flu), Common Cold (Upper Respiratory Tract Infection)
Spider angiomas on skin	Liver Cirrhosis
Sudden confusion or trouble understanding	Stroke (Cerebrovascular Accident)
Sudden dizziness, loss of balance or coordination	Stroke (Cerebrovascular Accident)
Sudden numbness or weakness of face/arm/leg	Stroke (Cerebrovascular Accident)
Sudden severe headache 'thunderclap'	Stroke (Cerebrovascular Accident)
Sudden significant weight gain from edema	Preeclampsia
Sudden vision changes in one or both eyes	Stroke (Cerebrovascular Accident)
Sudden, severe joint pain	Gout
Swan-neck and boutonniere deformities in advanced	Rheumatoid Arthritis
Sweating and heat intolerance	Hyperthyroidism
Swelling around joints	Osteoarthritis
Swelling of affected leg	Deep Vein Thrombosis (DVT)
Swelling of feet and ankles	Chronic Kidney Disease (CKD)
Swollen lymph nodes in axilla	Breast Cancer
Symmetric joint inflammation	Rheumatoid Arthritis
Symptoms worsen with stress and food	Irritable Bowel Syndrome (IBS)
Systemic features	Rheumatoid Arthritis
Tachycardia	Iron Deficiency Anemia
Thickened, cracked, or scaly skin	Atopic Dermatitis (Eczema)
Thickened, pitted, or ridged nails	Psoriasis
Throbbing or pulsating unilateral headache	Migraine
Tingling or numbness in hands/feet	Type 2 Diabetes Mellitus
Tophi	Gout

Tremor	Hyperthyroidism
Tunnel vision in advanced disease	Glaucoma
Unexplained iron-deficiency anemia	Colorectal Cancer
Unexplained weight loss	Type 2 Diabetes Mellitus, Peptic Ulcer Disease, Colorectal Cancer
Unintentional weight loss	Pulmonary Tuberculosis, Hyperthyroidism
Unusual cravings for non-nutritive substances	Iron Deficiency Anemia
Upper abdominal or epigastric pain	Preeclampsia
Uremic symptoms in advanced CKD	Chronic Kidney Disease (CKD)
Usually asymptomatic	Hypertension (High Blood Pressure)
Visual disturbances	Preeclampsia
Vomiting and diarrhea in young children	Acute Otitis Media (Middle Ear Infection)
Vomiting blood	Peptic Ulcer Disease
Warning signs of severe dengue	Dengue Fever
Watery eyes	Common Cold (Upper Respiratory Tract Infection)
Weight gain and difficulty losing weight	Polycystic Ovary Syndrome (PCOS)
Weight gain despite reduced appetite	Hypothyroidism
Weight loss in advanced disease	Chronic Obstructive Pulmonary Disease (COPD)
Wheezing	Asthma, Chronic Obstructive Pulmonary Disease (COPD)
Worsening after meals or when lying down	Gastroesophageal Reflux Disease (GERD)

Important Disclaimer

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